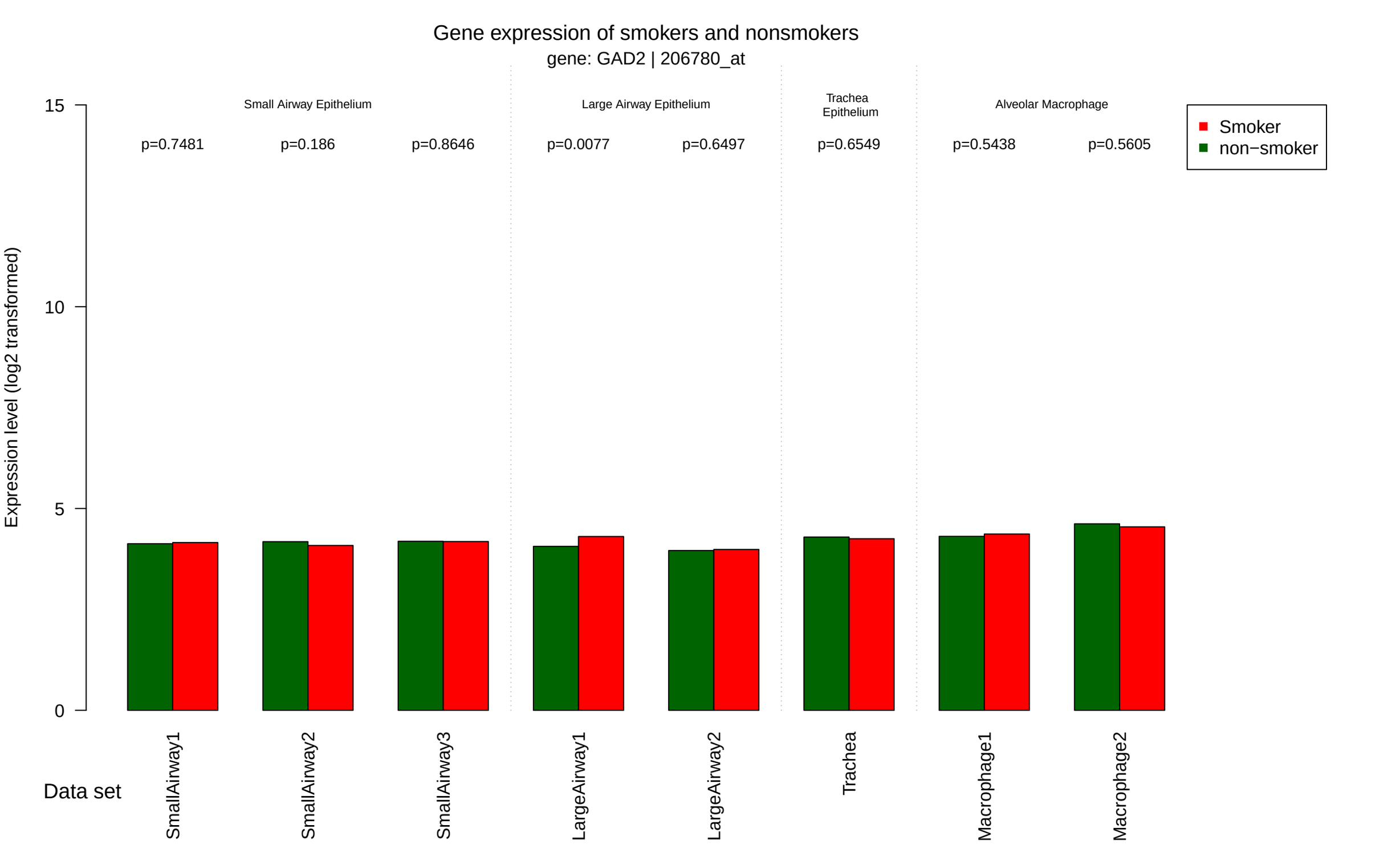
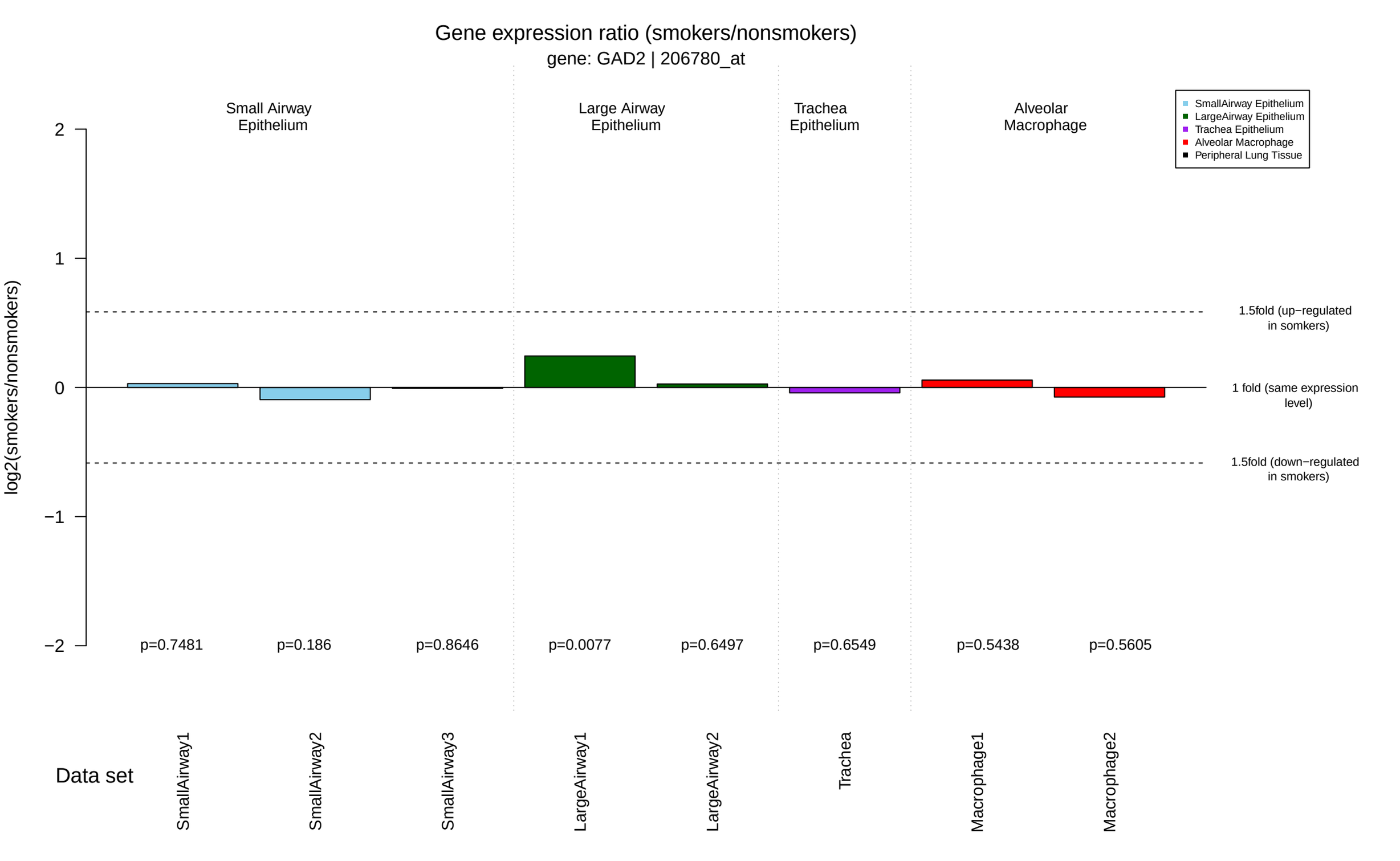
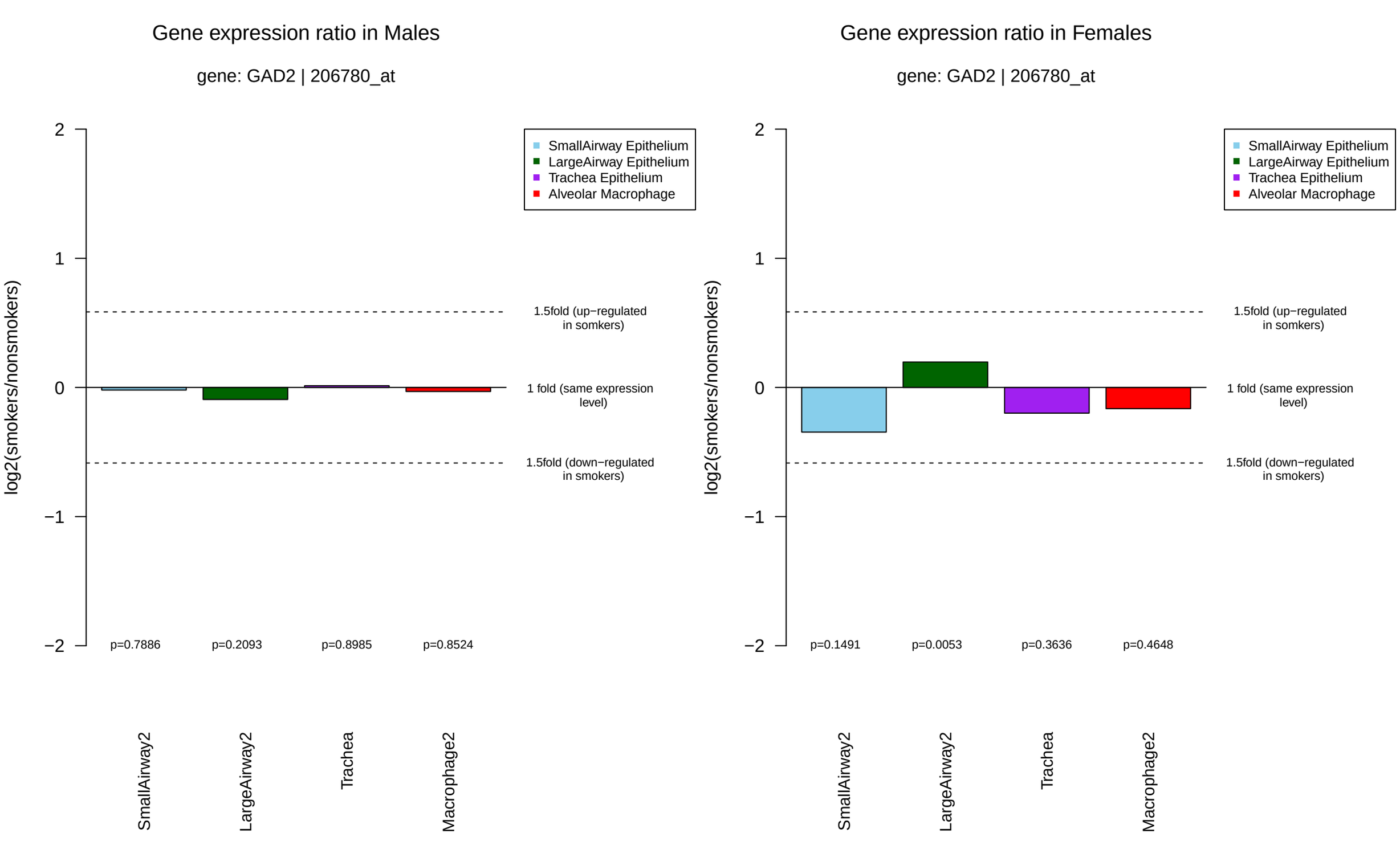
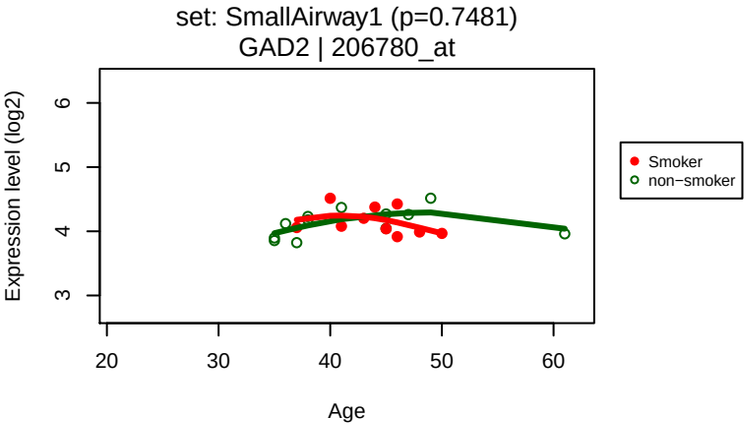
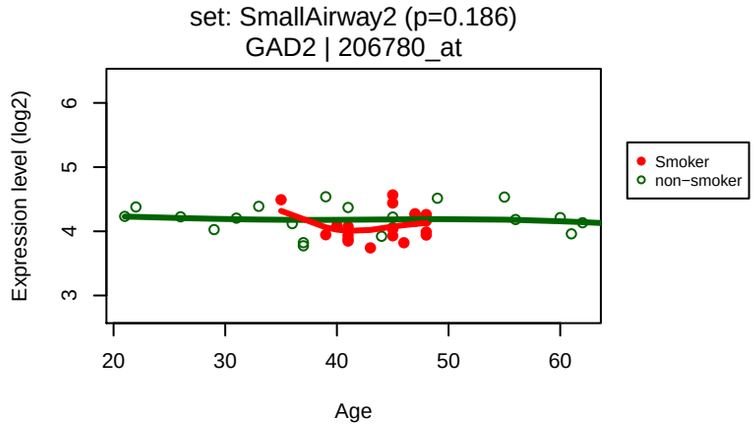
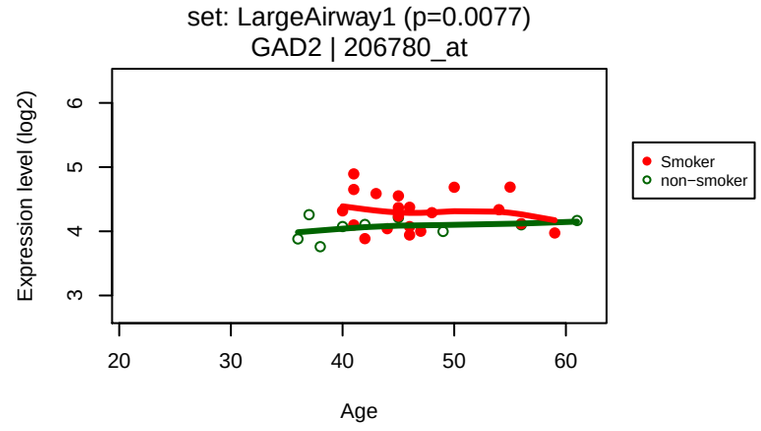

Gene expression of smokers and nonsmokers
gene: GAD2 | 206780_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7481
p=0.186
p=0.8646
p=0.0077
p=0.6497
p=0.6549
p=0.5438
p=0.5605
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

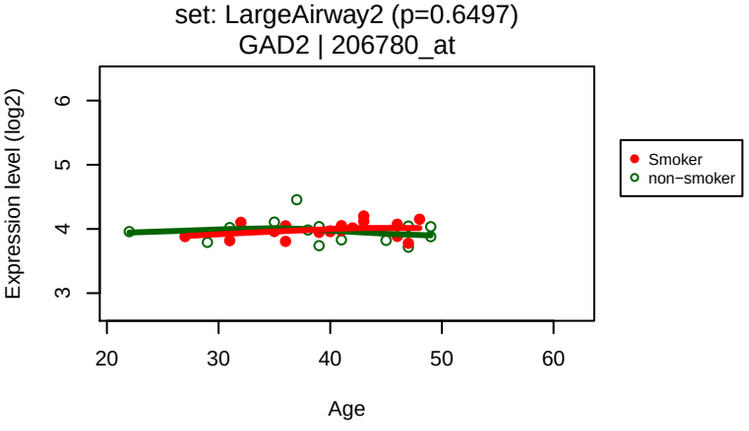

Gene expression ratio (smokers/nonsmokers)
gene: GAD2 | 206780_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7481
p=0.186
p=0.8646
p=0.0077
p=0.6497
p=0.6549
p=0.5438
p=0.5605
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

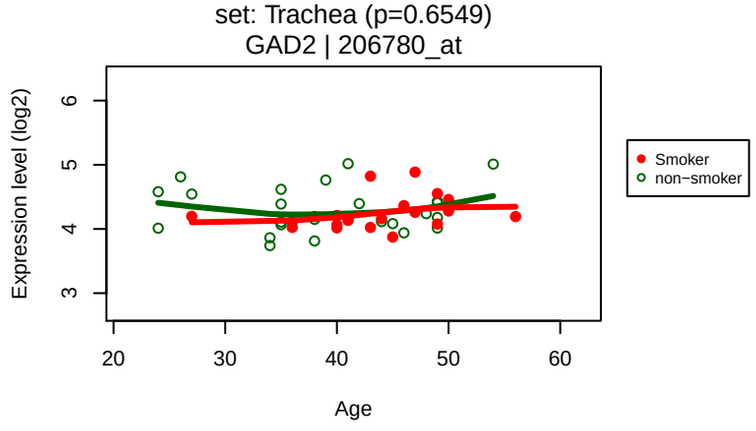

Gene expression ratio in Males
gene: GAD2 | 206780_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7886
p=0.2093
p=0.8985
p=0.8524
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAD2 | 206780_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1491
p=0.0053
p=0.3636
p=0.4648
SmallAirway2
LargeAirway2
Trachea
Macrophage2

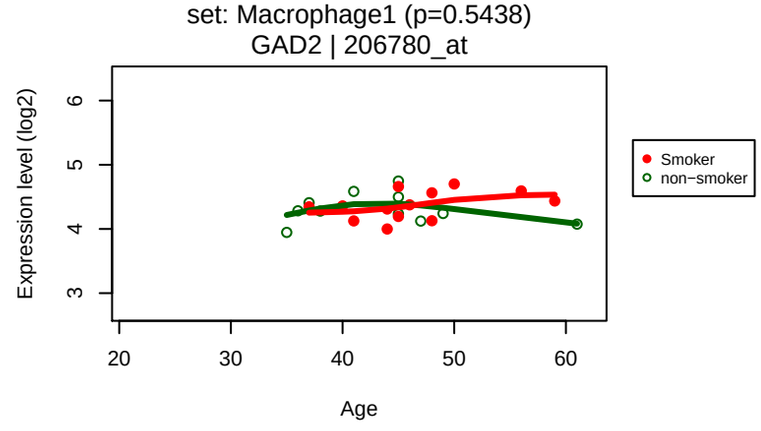

set: SmallAirway1 (p=0.7481)
GAD2 | 206780_at
Expression level (log2)
Age
Smoker
non−smoker

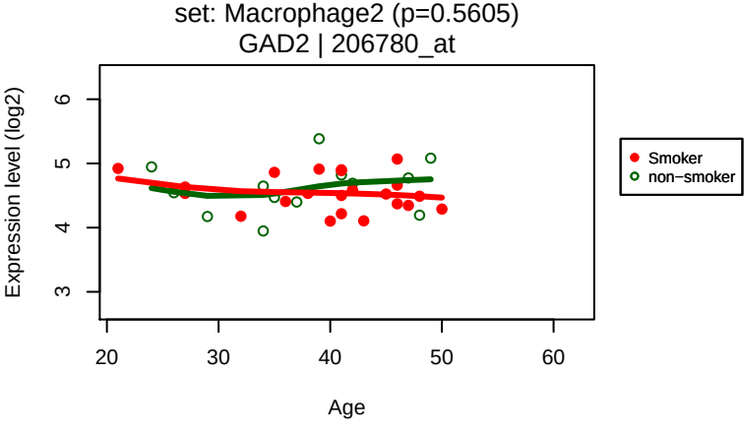

set: SmallAirway2 (p=0.186)
GAD2 | 206780_at
Expression level (log2)
Age
Smoker
non−smoker

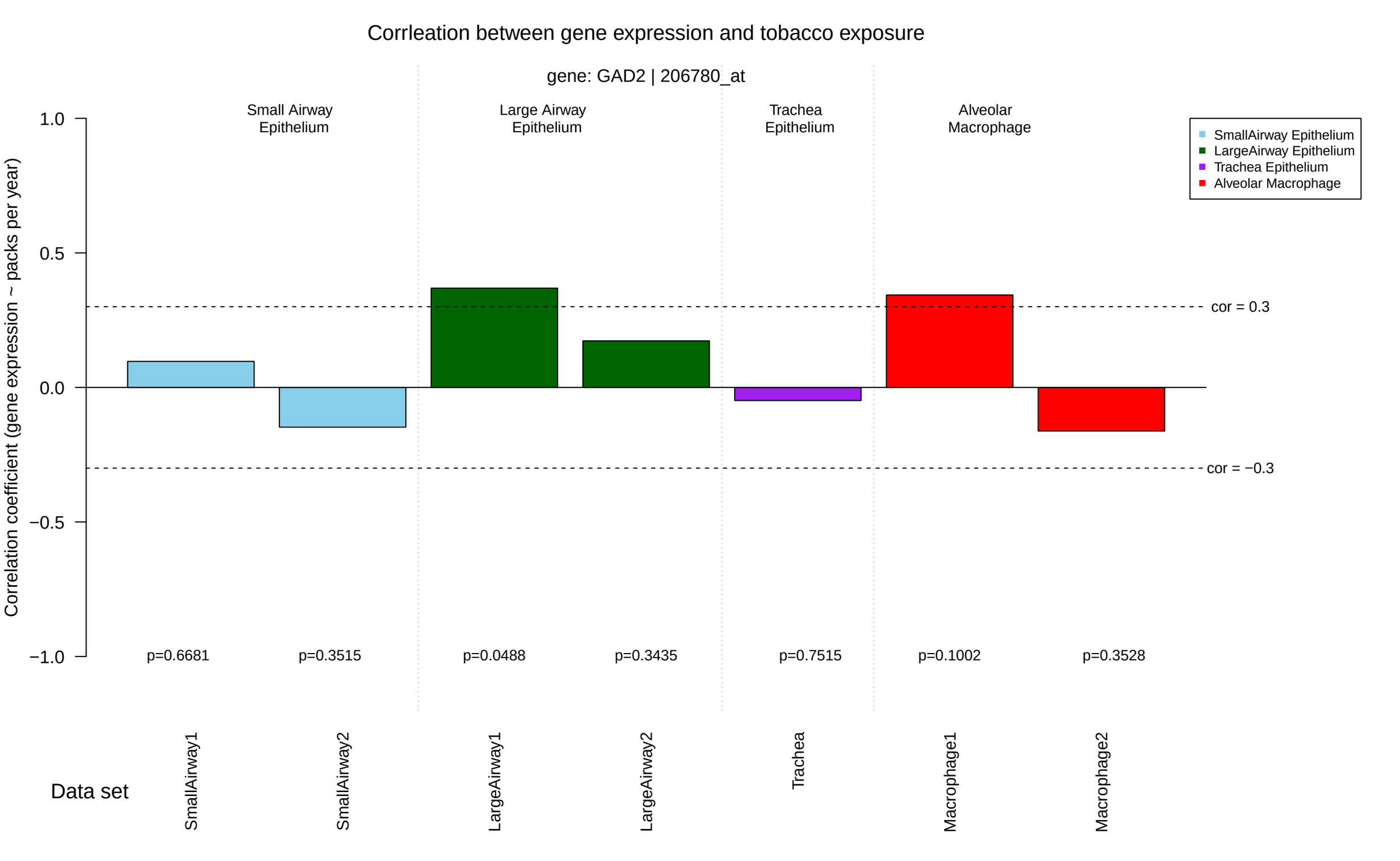

set: LargeAirway1 (p=0.0077)
GAD2 | 206780_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6497)
GAD2 | 206780_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6549)
GAD2 | 206780_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5438)
GAD2 | 206780_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5605)
GAD2 | 206780_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAD2 | 206780_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6681
p=0.3515
p=0.0488
p=0.3435
p=0.7515
p=0.1002
p=0.3528
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2